

Valid Context Window: Real-world Translation Task Evaluation Study based of OFAT Raw Text Data and Open Source LLMs

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Open source “Over Flow And Throw” Chinese & English raw text data in GitHub repository :

y-in-gb/open-fiction-access-token

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Background

Large Language Models (LLMs) with long context window show escalating hallucination and bad memory (e.g. forget to follow instruction) issues as the input gets longer. Although the severity and task performance varies for different open source LLM providers and model size, these issues appear commonly in many well-known open source LLMs.

Abstract

This study aims to provide reproducible experiment methodology and results to introduce the idea of “**Valid Context Window**” through a real-world task of translating the fiction “**Over Flow And Throw**” (OFAT) from Chinese to English.

Definition of Context Window

The idea of “Valid Context Window” is built upon “Context Window” also known as “Context Length,” which refers to the total combined token capacity of a single LLM inference:

Context Window = Input tokens (Total Message Tokens) + Output tokens (LLM Generated Tokens)

Definition of Total Message Tokens

Total Message Tokens = System Prompt tokens + User Prompt tokens (Input Chinese Text Tokens)

Here’s an example process of converting input text to tokens:

Input Text [Chinese text with 602 **characters**] → Model Tokenizer → Input Tokens [592 **tokens**]

Every LLM has its own model tokenizer, so it’s not always the case that the number of tokens will be less than the number of characters after tokenization, it can be larger, and it varies depend on the model tokenizer.

Definition of Valid Context Window

“**Valid**” means the LLM generated output result (e.g. the English translation) is good enough, it does not need to be perfect, but just good enough so that 0 human effort (e.g. editing) is required to amend the result.

“**Valid Context Window**” (VCW) is the context window that can allow the LLM to generate purely valid output result for a specific task, and in this study, it’ll be the Chinese to English translation task.

Since the length of the output English translation is dictated by the input Chinese text length, the output tokens(which does not contain system prompt) will be relatively stable and always be smaller than the input tokens(which contain system prompt).

Therefore, the only variable required to manipulate is the input tokens. The input tokens that allow LLM to generate valid output will be refer as the “**Valid Input Tokens**” (VIT).

Why “Valid” is so important?

Because the editing experience produced upon the output is hard to accumulate and reuse in any other LLMs. Flaws like wrong choosing of word in the output can be subtle, but the problem is those flaws cannot be 100 percent taken care by regex. LLMs can sprout all kinds of different word choices and combinations to make the resulting sentence not smooth, coherent, or make sense to human.

Instead of fixing endless small flaws from the output side, I think it’s more worthy to try fixing it from source, which is the input. I can build a vocabulary set to past on word knowledge as the context to any LLMs as long as they have the basic ability to follow instruction.

Task-related knowledge should be easily portable as context for any LLM to facilitate their task. My thought is to explore a new way for the LLMs to adapt just enough knowledge, and then generate valid output, but I will need to design another set of experiments to consolidate my hypothesis by testing out the most LLM adaptable format of the vocabulary set in future study.

Goal

In the current study, my goal is to test out the VIT for various LLMs, and within the VIT, the LLMs are expected to have 0 hallucination, 0 bad memory issue, and able to generate valid results.

Quick Look of the Final Results

Llama, Mistral, Qwen, and GLM models with context window of 32k or 128k are tested in this study, and surprisingly, the empirical VCW for Chinese to English translation task is much smaller than the model’s context window.

A model with a 32k context window had a measured VCW of around 4k tokens.

A model with a 128k context window also had a measured VCW of around 4k tokens.

This suggest a huge gap exist between theoretical capacity and real-world utility. While these models can technically ingest 32k+ tokens, their VCW for reliable translation consistently falls within just 3% to 12% of the context window.

Methodology

Experiment Set Up

Access to computing resource with GPU and at least 16 GB of RAM

Source Data

OFAT Chinese & English raw text content

[GitHub repository: <https://github.com/y-in-gb/open-fiction-access-token>]

Experiment Prompt Structure

1. System prompt

This prompt itself is already a stand-alone **Chinese to English AI Translation Agent**. It consists of role definition, instruction to follow given context (which contains few-shot translation examples), output format, and instruction to start the translation.

2. User prompt

Only the **Input Chinese Text** needed to be translated, nothing else.

Experiment Roadmap

1. Identify all the important features an LLM need to satisfy, so that it will have the potential to adapt word knowledge on the fly.
2. Find the VIT and VCW for each LLM.

Experiment Evaluation/Grade Point for LLMs

<i>Evaluation/Grade Point</i>	<i>Insights Behind the Grading</i>	<i>Expected Failure (Point Loss)</i>
Can generate fluent and human-readable English	Test whether the LLM’s pre-training data and method enable this LLM to satisfy the grade point	Unnatural language generation
Language Understanding: - No inappropriate literal translation - Know enough commonly-used English words	Test whether the LLM’s pre-training data and method enable this LLM to satisfy the grade point	- Inappropriate literal translation - Inappropriate literal translation for commonly-used words
Able to follow everything in the system prompt and always remember instruction to follow the given context	Test whether the LLM’s instruction fine-tuning data and method enable this LLM to satisfy the grade point	Bad memory issues: - Forget to follow instruction - Forget to translate some of the sentences
No hallucination	Test the LLM as a black-box and observe the pattern to study when and how the hallucination will appear	Hallucination

Table 1. Experiment Evaluation/Grade Point Table

Experiment 1

Data Preparation

1. Input few-shot examples for System Prompt (examples_text):

Take the first 20 lines of Chinese & English text content from OFAT chapter 3

```
Example 1 Chinese: 第三章：篡
Example 1 English: Chapter 3: Overthrow
...
Example 20 Chinese: ...
Example 20 English: ...
```

2. Input Chinese text for User Prompt:

Parse the remaining Chinese text content of OFAT chapter 6 (which are the content after OFAT-zh-chap4-6-chunk8 and within OFAT-zh-chap4-6) **line by line**

Result: 85 lines of text

System Prompt Hierarchical Template

1. examples_text consist of a list of example strings where each string is formatted as shown below.

```
f"Example {example_index} Chinese: {zh}\nExample {example_index}\nEnglish: {en}"
```

2. examples_section specify the instructions that the LLM need to follow during translation and provide necessary context (**examples_text**) to the LLM.

```
examples_section = f"""Follow the style, tone, and vocabulary choices demonstrated in the example below:
```

```
{examples_text}
```

Important guidelines:

- Maintain the same level of formality as the example
- Use similar vocabulary choices and expressions
- Preserve the original meaning while adapting to natural English
- Keep the same sentence structure patterns where appropriate"""

3. system_prompt states the role definition, output format, and instruction to begin the translation.

```
system_prompt = f"""You are a professional translator specializing in Chinese to English translation.
```

```
{examples_section}
```

Translate the following Chinese text to English following the example's style. Output ONLY the translated English text, nothing else, which means no explanations, no notes, no labels, just the translation itself.

```
Chinese text to translate: """
```

User Prompt

1 line of Input Chinese text

Model Hyperparameters

- temperature = 0.3 (for all LLMs)
- max_tokens = 5000 (for all LLMs)
- disable the thinking mode (for any LLMs with thinking mode on as default)

Experiment 1 – Overall Results

You may refer to the `token_cost_for_each_model_exp1.md` or `html` file in the GitHub repository of this research to check out the **Line of Text (L1 to L85)**, System Prompt Text Length (chars), System Prompt Tokens, Input Chinese Text Length (chars), Input Chinese Text Tokens, Total Message Tokens for Experiment 1.

The exact failure portions are highlighted with Underline as shown in the “**Failure Output Results & Descriptions**” column.

Models	Failures	Failure Output Results & Descriptions
meta-llama/ Llama-3.2- 3B-Instruct [1] (4 bit)	- Forget to follow instruction - Forget to translate some of the sentences - Hallucination	Forget to translate the 2nd sentence in the input line of text and generated extra note after the translation result e.g. “地牢。”手臂快要没有知觉了。The prison. <u>(Note: The translation maintains the same level of formality as the original Chinese text, using similar vocabulary choices and expressions, and preserving the original meaning while adapting to natural English.)</u> Forget to translate some of the sentence and Context Hallucination (the generated result is greatly hallucinated by the format of the 20 few-shot examples given in context when there should only be 1 output sentence for the translation) e.g. 我。。。我不知道。I'm so confused. <u>Example 1 English: I'm so confused.</u> <u>Example 2 English: I don't know.</u> ... <u>Example 20 English: I'm at a loss.</u> Model that has 7B parameters or above does not have these kind of Failure Output Results. Hence, the later experiments will mainly focus on testing 7B, 8B, and 9B model.
meta-llama/ Llama-3.1- 8B-Instruct [2] (4 bit)	Hallucination	Subject Hallucination e.g. 你和我的血脉会完美继承我们的优点。 <u>He and I will have a perfect bloodline, inheriting our best traits.</u> Context Hallucination e.g. 大雨下了一晚，洗刷着一切。<u>Chapter 1: Purification</u> Experiment 2 fixed the above issues by providing longer input text (from 1 line to 1 paragraph).

mistralai/ Mistral- Nemo- Instruct-2407 [3] (4 bit)	- Inappropriate literal translation for commonly-used words - Hallucination - Forget to translate some of the sentences	Translate commonly-used Chinese word to pinyin e.g. 不。不。 "Bu. Bu." - Subject Hallucination - Missing some subject and words e.g. 你和我的血脉会完美继承我们的优点。 <u>He</u> will inherit our virtues perfectly. Experiment 2 fixed the above issues by providing longer input text (from 1 line to 1 paragraph).
Qwen/ Qwen3-4B- Instruct-2507 [4] (8 bit)	- Unnatural language generation - Inappropriate literal translation for commonly-used words	Wrong use of “;” e.g. 楚寒不语，他就当是默认了。 Chu Han said nothing; he simply treated it as a tacit acceptance. Qwen3-8B does not have this issue. “小腿” is split to “小”，“腿”， and then translated to “<u>small leg</u>” instead of “calf” Qwen3.5-9B does not have this issue but the gpu computation is much more heavy.
Qwen/ Qwen3-8B [4] (4 bit)	Hallucination	Sentence such as "Chu Han, why did you go to me that day?" keep repeating in every translation starting from the 10th line (L10) of input text, all the way to the 85th line (L85) e.g. 10th line (L10) “去作甚？”<u>"Xue Niaoer looked at Chu Han, asking, 'Chu Han, why did you go to me that day?'"</u> 20th line (L20) 我。。。我不知道。<u>Xue Niaoer looked at Chu Han, "Chu Han, why did you go after me that day? No. I... I don't know."</u> 30th line (L30) 交给我。我也是你的一部分。我不会骗你的。<u>Xue Niaoer's eyes said, "Chu Han, why did you go after me that day? No. I... I don't know. Leave it to me. I am also part of you. I will not deceive you.</u> Qwen3-4B-Instruct-2507, Qwen2.5-7B-Instruct, Qwen2.5-7B-Instruct-1M, and Qwen3.5-9B do not have this issue. Experiment 2 fixed this issue by providing longer input text (from 1 line to 1 paragraph).

zai-org/ glm-4- 9b-0414 [10] (4 bit)	Inappropriate literal translation for commonly-used words	Left out Chinese word in English translation result e.g. “惨叫” is translated to “惨叫”
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Table 2. Experiment 1 overall results and descriptions

Experiment 1 – Bloodline Test Results

An interesting finding worth to highlight in Experiment 1 is the translation of the first sentence in the **55th line (L55)** Input Chinese Text: “你和我的血脉会完美继承我们的优点。” I will referred to this sentence as the bloodline test in later experiments. I found this test very useful for testing LLM’s “Hallucination” and “Unnatural language generation” failures, you will see why as I repeat the bloodline test in Experiment 2.

To break it down a little more, this test is actually answering the question of what makes language “natural” in the first place.

From the Chinese side, it will be unnatural to convey “你和我的” as “你的和我的”.

From the English side, LLMs translate “你和我的” as shown below:

Models	Translation Results	Failures
meta-llama/Llama-3.2-3B-Instruct [1] (4 bit)	<u>You and my</u> bloodline will inherit our perfect virtues.	Inappropriate literal translation “你和我的” is split to “你”, “和”, “我的”, and then translated to “You”, “and”, “my” correspondingly
meta-llama/Llama-3.1-8B-Instruct [2] (4 bit)	<u>He</u> and I will have a perfect bloodline, inheriting our best traits.	Hallucination Subject Hallucination
mistralai/Mistral-Nemo-Instruct-2407 [3] (4 bit)	<u>He</u> will inherit our virtues perfectly.	Hallucination Subject Hallucination
Qwen/Qwen3-4B-Instruct-2507 [4] (8 bit)	<u>Your</u> bloodline will perfectly inherit our strengths.	Hallucination Subject Hallucination
Qwen/Qwen3-8B [4] (4 bit)	<u>Chu Han, why did you go after me that day? No. Because you are a monster... Okay. Truly, it's so boring.</u> "Bi Xia, your teeth and my bloodline will perfectly inherit our virtues.	Hallucination Context Hallucination

Qwen/Qwen2.5-7B-Instruct [5][6] (4 bit)	<u>You and my</u> bloodline will perfectly inherit our 优点。	- Inappropriate literal translation “你和我的” is split to “你”, “和”, “我的”, and then translated to “You”, “and”, “my” correspondingly
Qwen/Qwen2.5-7B-Instruct-1M [7][8] (4 bit)	<u>You and my</u> bloodline will perfectly inherit our 优点。	- Inappropriate literal translation for commonly-used words Left out Chinese word in English translation result
Qwen/Qwen3.5-9B [9] (4 bit)	<u>You and my</u> bloodline will perfectly inherit our strengths.	Inappropriate literal translation “你和我的” is split to “你”, “和”, “我的”, and then translated to “You”, “and”, “my” correspondingly
zai-org/glm-4-9b-0414 [10] (4 bit)	Your bloodline with mine will perfectly inherit our virtues.	0 failure

Table 3. Experiment 1 bloodline test results

This test shows evidences supporting that “LLMs can sprout all kinds of different word choices and combinations to make the resulting sentence not smooth, coherent, or make sense to human.”

One more thing to note is that, the rest of the sentences left in the **55th line (L55)** of Input Chinese Text contains a bunch of four-character idiom that need to be defined in the vocabulary set. It can be helpful to observe how well the LLMs be able to adapt the vocabulary knowledge and make the final translation for this whole line of Input Chinese Text fluent in future study.

Experiment 1 – Results Observation & Direction for Experiment 2

Based of the results in **Table 2** and **Table 3**, I found “Inappropriate literal translation” failures appear to be as common as “Hallucination” and “Bad memory” issues defined in **Table 1**.

If a LLM does not perform well on a task, a common sense solution is to provide more context to the LLM, and this approach will be what I want to test in Experiment 2. By augmenting the Input Chinese Text, will each types of failures in **Table 2** and **Table 3** disappear or deteriorate.

Experiment 2

Experiment 2 is developed based on Experiment 1 with just two modifications as shown below:

1. Remove the word “第三章： ” and “Chapter 3:” from the **examples_text**.
2. Modify “Output ONLY the translated English text, nothing else, which means no explanations, no notes, no labels, just the translation itself” to “Output ONLY the translated English text and nothing else” in the **system_prompt**.

These two minor modifications aim to eliminate the input context (especially the few-shot examples) sensitivity for Llama-3.1-8B-Instruct.

Data Preparation

1. Input few-shot examples for System Prompt (examples_text):

Take the first 20 lines of Chinese & English text content from OFAT chapter 3

```
Example 1 Chinese: 篡
Example 1 English: Overthrow
...
Example 20 Chinese: ...
Example 20 English: ...
```

2. Input Chinese text for User Prompt:

Parse the remaining Chinese text content of OFAT chapter 6 (which are the content after OFAT-zh-chap4-6-chunk8 and within OFAT-zh-chap4-6) **paragraph by paragraph** using:

```
split('\n\n')
```

Result: 5 paragraphs

System Prompt Hierarchical Template

1. examples_text

Same as Experiment 1

2. examples_section

Same as Experiment 1

3. system_prompt

```
system_prompt = f"""You are a professional translator specializing in
Chinese to English translation.
```

```
{examples_section}
```

```
Translate the following Chinese text to English following the example's
style. Output ONLY the translated English text and nothing else.
```

```
Chinese text to translate: """
```

User Prompt

1 paragraph of Input Chinese text

Model Hyperparameters

Same as Experiment 1

Experiment 2 – Overall Results

You may refer to the `token_cost_for_each_model_exp2.md` or `html` file in the GitHub repository of this research to check out the **Paragraph (P1 to P5)**, System Prompt Text Length (chars), System Prompt

Tokens, Input Chinese Text Length (chars), Input Chinese Text Tokens, Total Message Tokens for Experiment 2.

P1 (L1 – L2), P2 (L3 – L31), P3 (L32 – L43), P4 (L44 – L50), P5 (L51 – L85)

The exact failure portions are highlighted with Underline.

<i>Models</i>	<i>Failures</i>	<i>Failure Output Results, Descriptions, and Improvements</i>
meta-llama/ Llama-3.1- 8B-Instruct [2] (4 bit)	P1: 0 failure P2: - Unnatural language generation - Hallucination P3: 0 failure P4: - Unnatural language generation P5: - Hallucination - Forget to translate some of the sentences - Inappropriate literal translation for commonly-used words - Forget to follow instruction	P2 e.g. “<u>逼供出你的去向，松手！</u>” "Trying to extract information about your whereabouts(Unnatural language generation), <u>and release them(Hallucination).</u>" <u>Chu Han replied, his forehead dripping with sweat.(Context Hallucination)</u> The same model in Experiment 1 got: "Forcefully <u>extract</u> (Unnatural language generation) your whereabouts <u>from you(Hallucination)</u>, let go!" P3 Improvements “真是无趣。” is translated to “How boring.” The same model in Experiment 1 got: “<u>It's truly uninteresting.</u>” (Inappropriate literal translation) P4 e.g. “你和她怎的都不理人？” "How can you two <u>not even bother</u>(Unnatural language generation) with me?" The same model in Experiment 1 got: "How can you two <u>not care</u>(Unnatural language generation) for <u>people</u>(Subject Hallucination)?" P5 e.g. - “Chu Han gasped for breath,” is translated to “<u>He</u> gasped for breath,” (Subject Hallucination. The same model in Experiment 1 does not have this issue) - “Chu Han turned his head to avoid it, <u>but he was caught by the emperor</u>” (Subject Hallucination since Chu Han is the emperor. The same model in Experiment 1 also have this issue but not as worse. Forget to translate some of the sentences) - “<u>small leg</u>”(Inappropriate literal translation for commonly-used words. The same model in Experiment 1 also have this issue) - “Bi Xia” is translated to “<u>My emperor</u>”(Forget to follow instruction. The same model in Experiment 1 does not have this issue)

mistralai/ Mistral- Nemo- Instruct-2407 [3] (4 bit)	P1: 0 failure P2: - Forget to translate some of the sentences - Forget to follow instruction P3: Hallucination P4: Hallucination P5: - Hallucination - Forget to follow instruction - Forget to translate some of the sentences P2 and P4 loss the newline structure of the original content by concatenating all sentences without any newline after translation.	P2 failures e.g. - “脸色开始发白” is translated to “face pale”(Forget to translate some of the sentences. The same model in Experiment 1 does not have this issue) - “雪裊儿” is translated to “<u>Snow</u> Niaoer” instead of “Xue Niaoer” given in the few-shot examples (Forget to follow instruction.The same model in Experiment 1 does not have this issue) P2 improvements Majority of the sentences in P2 are one-sentence dialogue, and by augmenting Input Chinese Text from 1 line to 1 paragraph, this model is able to translate much better than in Experiment 1. (e.g. “不。不。” is translated to “No. No.”) P3 failures e.g. “雪裊儿的眼睛” is translated to “<u>Her</u> irises” (Subject Hallucination. The same model in Experiment 1 does not have this issue, but “眼睛” is falsely translated to “<u>pupils</u>” in Experiment 1) P3 Improvements “真是无趣。” is translated to “How dull.” The same model in Experiment 1 got: “<u>That's quite dull.</u>” (Unnatural language generation) P4 e.g. “咬” is translated to “<u>held</u>” (Hallucination. The same model in Experiment 1 does not have this issue) P5 e.g. - “楚寒” is translated to “<u>her</u>” (Subject Hallucination. The same model in Experiment 1 does not have this issue) - “陛下” is translated to “<u>Your Majesty</u>” instead of “Bi Xia”(Forget to follow instruction) - Forget to translate part of the sentence at L58 in P5 (The same model in Experiment 1 does not have this issue) - Forget to translate some of the sentences at L81 in P5 (The same model in Experiment 1 does not have this issue)
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Qwen/ Qwen2.5-7B-Instruct [5][6] (4 bit)	P1: 0 failure P2: - Inappropriate literal translation - Unnatural language generation - Forget to translate some of the sentences - Hallucination P3: - Inappropriate literal translation P4: - Forget to translate some of the sentences - Hallucination P5: - Forget to translate some of the sentences - Hallucination - Forget to follow instruction - Inappropriate literal translation	P2 e.g. - “烙铁” is translated to “烙 iron” (Inappropriate literal translation. The same model in Experiment 1 does not have this issue) - “逼供出你的去向，松手！” "To <u>extract</u> your <u>location</u>, <u>release me</u>!" (Unnatural language generation) The same model in Experiment 1 got: "Force <u>you</u> (Subject Hallucination) to reveal your whereabouts, <u>then</u> (Unnatural language generation) let go!" - Forget to translate the dialogue at L14 in P2 (Forget to translate some of the sentences. The same model in Experiment 1 does not have this issue) - “by him” is translated to “by you” (Subject Hallucination. The same model in Experiment 1 does not have this issue) P3 e.g. “真是无趣。” is translated to “<u>Truly boring.</u>” (Inappropriate literal translation) The same model in Experiment 1 got: “<u>It's truly boring.</u>” (Inappropriate literal translation) P4 e.g. - Forget to translate part of the sentence at L44 in P4 (Forget to translate some of the sentences. The same model in Experiment 1 does not have this issue) - “你和她怎的都不理人？” "Why are you and her ignoring <u>everyone</u>?" (Subject Hallucination) The same model in Experiment 1 got: "Why are you both ignoring <u>people</u>?" (Subject Hallucination) P5 e.g. - Forget to translate part of the sentence at L58 in P5 (The same model in Experiment 1 does not have this issue) - Mixed up the cause-and-effect for “illness” at L64 in P5 (Hallucination. The same model in Experiment 1 does not have this issue) - “陛下” is translated to “<u>His Majesty</u>” instead of “Bi Xia” (Forget to follow instruction. The same model in Experiment 1 does not have this issue) - “稍不留神瞄了一眼” is translated to “<u>not looking</u>” (Inappropriate literal translation. “不”，“瞄” are translated to “not”，“looking” correspondingly) The same model in Experiment 1 got: “barely glanced”
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zai-org/ glm-4- 9b-0414 [10] (4 bit)	P1: 0 failure P2: - Hallucination - Inappropriate literal translation P3: - Inappropriate literal translation P4: 0 failure	P2 improvement “逼供出你的去向，松手！” "To force an confession about your whereabouts, loosen your hand!" The same model in Experiment 1 got: "Force <u>me</u> (Subject Hallucination)to reveal where you're going, let go!" P2 e.g. - “供出来了吗？” "Have <u>you</u> confessed?" (Subject Hallucination. The same model in Experiment 1 also have this issue) - “她是人” “she <u>was</u> a <u>person</u>” (Inappropriate literal translation) The same model in Experiment 1 got valid translation: “she is a human” - “她永远都回不来了，你甘心吗？” “she’ll never come back. <u>Are you willing?</u>” (Inappropriate literal translation) The same model in Experiment 1 got: “will you be content <u>if</u> (Hallucination) she can never return?” P3 e.g. - “幽蓝色” is translated to “幽蓝色” (Inappropriate literal translation. The same model in Experiment 1 does not have this issue) - “包裹” is translated to “包裹” (Inappropriate literal translation. The same model in Experiment 1 also have this issue and translated the word to “包裹物”) - “真是无趣。” is translated to “It’s truly boring.” (Inappropriate literal translation. The same model in Experiment 1 also have this issue) - “凭感觉” is translated to “by feel” (Inappropriate literal translation. The same model in Experiment 1 also have this issue)
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	P5: - Hallucination - Inappropriate literal translation - Unnatural language generation P3 loss the newline structure of the original content by concatenating all sentences without any newline after translation.	P5 e.g. - “不得安宁” is translated to “in peace” at L52 in P5 (Hallucination. Missing negative word such as “not”. The same model in Experiment 1 does not have this issue at this line, but it have another Hallucination issue which missed the words “you” after translation) - Subject Hallucination at L53 in P5 (The same model in Experiment 1 does not have this issue) - “窝” is translated to “窟” (Inappropriate literal translation. The same model in Experiment 1 also have this issue) - “龙涎香” is translated to “loong-eneair” (Context Hallucination. “loong-eneair” is a custom-defined English vocabulary to translate “龙气” as given in few-shot examples context) The same model in Experiment 1 nicely adapt the context and translated the word as: “loong-scent” - “调和气息” is translated to “调和气息” and “陛下” is translated to “陛下” (Inappropriate literal translation. The same model in Experiment 1 does not have this issue) - Mixed up the cause-and-effect for “illness” at L64 in P5 (Hallucination. The same model in Experiment 1 does not have this issue) - “The <u>only way</u> to ... <u>is one</u>.” (Unnatural language generation. The same model in Experiment 1 also have this issue) - “小腿” is translated to “calves” or “legs” (Hallucination) - “<u>with pressed forward</u>” (Unnatural language generation. The same model in Experiment 1 does not have this issue)
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Table 4. Experiment 2 overall results and descriptions

Experiment 2 – Bloodline Test Results

<i>Models</i>	<i>Translation Results</i>	<i>Failures</i>
meta-llama/ Llama-3.1- 8B-Instruct [2] (4 bit)	Experiment 1: <u>He</u> and I will have a perfect bloodline, inheriting our best traits. Experiment 2: <u>You and my</u> bloodline will perfectly inherit our advantages.	Experiment 1: Hallucination Subject Hallucination Experiment 2: Inappropriate literal translation “你和我的” is split to “你”, “和”, “我的”, and then translated to “You”, “and”, “my” correspondingly
mistralai/ Mistral- Nemo- Instruct-2407 [3] (4 bit)	Experiment 1: <u>He</u> will inherit our virtues perfectly. Experiment 2: Our child will perfectly inherit our virtues.	Experiment 1: Hallucination Subject Hallucination Experiment 2: 0 failure
Qwen/ Qwen3-8B [4] (4 bit)	Experiment 1: <u>Chu Han, why did you go after me that day? No. Because you are a monster...</u> <u>Okay. Truly, it's so boring.</u> "Bi Xia, <u>your teeth and my</u> bloodline will perfectly inherit our virtues. Experiment 2: <u>You and my</u> bloodline will inherit our virtues perfectly.	Experiment 1: Hallucination Context Hallucination Experiment 2: Inappropriate literal translation “你和我的” is split to “你”, “和”, “我的”, and then translated to “You”, “and”, “my” correspondingly
Qwen/ Qwen2.5-7B- Instruct [5][6] (4 bit)	Experiment 1: <u>You and my</u> bloodline will perfectly inherit our 优点。 Experiment 2: <u>Your</u> bloodline will perfectly inherit our strengths.	Experiment 1: - Inappropriate literal translation “你和我的” is split to “你”, “和”, “我的”, and then translated to “You”, “and”, “my” correspondingly - Inappropriate literal translation for commonly-used words Left out Chinese word in English translation result Experiment 2: Hallucination Subject Hallucination or Forget to translate some of the sentences Missing “with mine” after translation
zai-org/glm-4- 9b-0414 [10] (4 bit)	Experiment 1: Your bloodline with mine will perfectly inherit our virtues. Experiment 2: <u>Your</u> bloodline will perfectly inherit our strengths.	Experiment 1: 0 failure Experiment 2: Hallucination Subject Hallucination or Forget to translate some of the sentences Missing “with mine” after translation

Table 5. Experiment 2 bloodline test results

In Experiment 2, glm-4-9b-0414 failed to generate valid output due to input tokens exceeding the model’s VIT, and caused hallucination or bad memory issues.

Note that the sentence for bloodline test falls under P5, which is the input paragraph that cause the most hallucination and bad memory issues out of all paragraphs for every LLM in Experiment 2.

The evaluation/grading experience I accumulated over the two experiments gifted me an inspiration is that, the process of evaluation and valid results collection can feed and loop back to one another. For the bloodline test itself, I have already collected three valid translations:

1. Llama-3.1-8B-Instruct: You and I will have a perfect bloodline, inheriting our best traits. (I manually change “He” to “You” while preserving the translated sentence’s structure)
2. Mistral-Nemo-Instruct-2407: Our child will perfectly inherit our virtues.
3. glm-4-9b-0414: Your bloodline with mine will perfectly inherit our virtues.

And there can be more valid results to explore for the bloodline test, so as any other Input Chinese Text’s vocabularies, sentences, lines, or paragraphs.

VIT and VCW based on Token Cost and Experiment Results

Similar to Context Window, VCW can be calculated by:

$$\text{VCW} = \text{VIT} + \text{Output tokens}$$

And since the output tokens will always be smaller than the input tokens, I will assume the output tokens equal to the input tokens (**Total Message Tokens**) to estimate the VCW:

$$\text{VCW} = \text{VIT} + \text{VIT} = 2 * \text{VIT}$$

<i>Models</i>	<i>context window (tokens)</i>	<i>VIT (tokens)</i>	<i>VCW (tokens)</i>
meta-llama/Llama-3.1-8B-Instruct [2] (4 bit)	128k (according to model card)	Between 1.9k and 2.2k	~4.4k
mistralai/Mistral-Nemo-Instruct-2407 [3] (4 bit)	128k (according to model card)	Between 2k and 2.1k	~4.2k
Qwen/Qwen3-8B [4] (4 bit)	32,768 natively and 131,072 tokens with YaRN (according to model card)	Between 1.7k and 1.9k	~3.8k
Qwen/Qwen2.5-7B-Instruct [5][6] (4 bit)	131,072 (according to model card)	Between 1.7k and 1.8k	~3.6k
zai-org/glm-4-9b-0414 [10] (4 bit)	32K -> 128K (according to GLM-4 GitHub)	Between 1.67k and 1.85k	~3.7k

Table 6. VIT and VCW for each LLM

Limitation

- Lack of resources to test larger LLM with size 10B to 20B, 20B to 30B, 30B to 70B, and MoE models with more than 100B parameters
- Haven't repeat the same set of experiments with different temperature and other hyperparameter

Reflection

While my current empirical scope only cover LLMs with 10B parameters or below, the shared architectural lineage across scales leads me to a critical concern:

If the limitations of LLM are unable be overcome by larger context window or larger parameters size (which require more training data and more resource cost for training), they will continue to cause similar kinds of issues and generate invalid results.

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GitHub repository links for the open source LLMs used in this research

- Llama-3.2-3B-Instruct: <https://github.com/meta-llama/llama3>
- Llama-3.1-8B-Instruct: <https://github.com/meta-llama/llama3>
- Mistral-Nemo-Instruct-2407: <https://github.com/mistralai/mistral-inference>
- Qwen3-4B-Instruct-2507: <https://github.com/QwenLM/Qwen3>
- Qwen3-8B: <https://github.com/QwenLM/Qwen3>
- Qwen2.5-7B-Instruct: <https://github.com/QwenLM/Qwen2.5>
- Qwen2.5-7B-Instruct-1M: <https://github.com/QwenLM/Qwen2.5>
- Qwen3.5-9B: <https://github.com/QwenLM/Qwen3.6>
- GLM-4-9B-0414: <https://github.com/zai-org/GLM-4>